

Adaptive CCTV Compression System Using Dynamic Region of Interest (ROI) Control

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Abstract

Modern CCTV surveillance networks generate enormous volumes of video data, straining network infrastructure and storage systems while demanding significant computational resources. Conventional fixed-bitrate compression methods apply uniform quality across entire frames, wasting bandwidth on static backgrounds while potentially degrading critical forensic details during security incidents. This paper presents an adaptive CCTV compression system that leverages YOLOv8-based dynamic Region of Interest (ROI) control with a risk-aware event engine to achieve intelligent bandwidth reduction. The system employs a three-tier architecture comprising an edge node for real-time object detection and adaptive encoding, a central server for stream management and metrics collection, and an interactive web dashboard for live monitoring and control. A multi-signal risk score combining object presence, motion area, time-of-day, and scene change magnitude drives a three-state surveillance machine (NORMAL, ALERT, CRITICAL) with hysteresis, automatically transitioning to near-lossless recording during critical incidents. A rolling circular buffer preserves 15 seconds of pre-event forensic context. Experimental results demonstrate bandwidth reduction of 45–65% while maintaining stable PSNR (30–35 dB), SSIM (0.85–0.95), and VMAF scores, with enhanced evidence preservation during critical events.

Keywords: Video surveillance, adaptive compression, YOLOv8, ROI, risk assessment, edge computing

1 Introduction

The rapid proliferation of CCTV surveillance systems in Bangladesh and globally has introduced significant technical and economic challenges. With over 1.6 million cameras installed nationwide and the smart surveillance market expanding at 50% annually [1], conventional systems continue to stream video at fixed high bitrates, leading to critical inefficiencies [2]. Existing CCTV systems waste 40–50% of available network capacity even when the vast majority of the scene remains static, lack dynamic adjustment of video quality based on motion or threat level, and cannot distinguish between routine monitoring and critical security incidents.

Recent advances in Region of Interest (ROI)-based video compression have demonstrated significant potential for bandwidth efficiency in surveillance systems [3, 4]. However, existing approaches are predominantly focused on analytics performance, simulation-based validation, or machine-centric optimization [5, 6]. Critically, no existing work simultaneously addresses real-time edge deployment with neural object detection, user-controlled adaptive compression with live visualization, risk-aware event-driven quality preservation, and comprehensive evaluation spanning signal quality, perceptual quality, and investigation usability metrics.

This paper presents a complete end-to-end adaptive CCTV compression system that addresses all four dimensions. The key contributions are:

1. **YOLOv8-based intelligent ROI detection:** Real-time object detection across 80 COCO classes with three-tier priority classification (HIGH for persons, MEDIUM for vehicles, LOW for other objects), enabling semantically aware compression allocation.
2. **Risk-aware event engine:** A multi-signal risk score combining object presence, motion area, time-of-day, and scene change magnitude drives a three-state surveillance machine with hysteresis and pre/post-event frame buffering for forensic evidence preservation.
3. **Multi-tier system architecture:** Edge node (Python + OpenCV + YOLOv8 + FFmpeg) for capture and adaptive encoding, Node.js server for stream management and metrics collection, and Next.js dashboard for real-time monitoring and interactive control.
4. **Comprehensive evaluation pipeline:** Automated benchmarking computing global, per-ROI, and background-only PSNR/SSIM alongside VMAF scores and detection accuracy retention.

2 System Architecture

The proposed system follows a three-tier architecture comprising the Edge Node, Central Server, and Dashboard, with a separate Evaluation Pipeline for automated benchmarking [7].

Fig. 1 System architecture: three-tier edge-server-dashboard design with evaluation pipeline

Figure 1: System architecture overview

2.1 Edge Node

The edge node is implemented in Python and runs continuously on a device co-located with the camera. It performs four primary functions [8]:

Frame Acquisition: Video frames are captured from a live camera source (webcam or IP camera) using OpenCV. Each frame is processed in real time at a configurable frame rate (5–24 FPS depending on the quality profile).

Object Detection: The YOLOv8n (nano) neural network performs object detection across 80 COCO classes [9]. Detections are classified into three priority tiers: HIGH (persons, class 0), MEDIUM (vehicles including cars, motorcycles, buses, trains, trucks), and LOW (all other detected objects) [10, 11, 12, 13]. Detection is performed at configurable intervals (default: every 3 frames) to balance accuracy and performance [14].

Temporal ROI Persistence: A TTL (Time-to-Live) system ensures ROI bounding boxes persist for 30 frames (~ 2 seconds at 15 FPS), eliminating visual flickering. A `merge_rois()` function using Intersection-over-Union (IoU) threshold ≥ 0.3 smoothly blends new detections into existing bounding boxes, providing temporally stable ROI boundaries [15].

Risk-Aware Event Engine: A `risk_score()` function computes a normalized risk value [0.0, 1.0] per frame by combining five independent signals [16]:

$$R = w_p \cdot N_p + w_v \cdot N_v + w_m \cdot M + w_t \cdot T + w_s \cdot S \quad (1)$$

where N_p is the count of HIGH-priority objects (+0.30 each), N_v is the count of MEDIUM-priority objects (+0.10 each), M is the motion area fraction (up to +0.25), T is the after-hours bonus (22:00–06:00, +0.20), and S is the scene change magnitude (up to +0.15). This score drives a three-state surveillance machine:

Table 1: Risk-driven surveillance state machine

State	Risk Range	BG Scale	BG Quality	ROI Quality
NORMAL	0.00–0.25	User-defined	User-defined	User-defined
ALERT	0.25–0.55	0.75	45	95
CRITICAL	0.55–1.00	1.0	88	100

Hysteresis is implemented with an 8-frame hold period before downgrading to prevent rapid state oscillation [17]. A rolling circular buffer (15 seconds $\times$ FPS) continuously stores original frames. When the state transitions to CRITICAL, pre-event frames are dumped to disk with full metadata, followed by 10 seconds of post-event recording [15].

Fig. 2 Three-state surveillance machine with hysteresis: NORMAL
→ ALERT → CRITICAL

Figure 2: Risk-driven state machine

Adaptive Compression and Encoding: Each frame is subdivided into foreground (ROI) and background (non-ROI) regions. ROI regions are encoded at high quality while the background is downsampled and aggressively compressed. The `CodecManager` class manages an FFmpeg subprocess supporting runtime codec switching between `libx264`, `libx265`, `libsvtav1`, and `hevc_nvenc` [18]. Adjustable parameters include Background Scale (resolution reduction), Background Quality (JPEG compression level), ROI Quality, and Frame Skipping (detection interval) [19].

Fig. 3 Adaptive compression pipeline: ROI detection, background
downscaling, and H.265 UDP streaming

Figure 3: Compression pipeline flow

2.2 Central Server

The Node.js server manages two Socket.IO namespaces: `/stream` (edge node pushes frames) and `/view` (dashboard pulls frames and sends controls) [5]. A UDP server monitors H.265 stream bitrate in real time. Automatic bandwidth adaptation computes total bandwidth every 3 seconds and sends reduced-quality control messages if thresholds are exceeded. Comprehensive CSV logs maintain records: `metrics_log.csv`, `control_log.csv`, and `motion_log.csv`. HTTP endpoints include `/control`, `/metrics`, `/sample`, `/reconstruct`, `/playback`, `/event`, and `/config`.

2.3 Dashboard

The Next.js dashboard features canvas-based live stream rendering with priority-coded ROI overlay (RED=HIGH, YELLOW=MEDIUM, GREEN=LOW), real-time control sliders (BG

Scale, BG Quality, ROI Quality, Detect Every N Frames), preset mode buttons (Low BW and High Quality), dual bandwidth display (Raw Socket.IO kbps and H.265 UDP kbps with % Saved), live Chart.js bandwidth chart, and color-coded state badge (NORMAL green, ALERT orange, CRITICAL red).

Fig. 4 Live dashboard: canvas stream, ROI overlays, control sliders, bandwidth chart, and state badge

Figure 4: Dashboard interface

3 Methodology

3.1 Frame Acquisition and Detection Pipeline

Video frames are captured using OpenCV and processed through the YOLOv8n detection model. The detection pipeline classifies objects into priority tiers and constructs ROI masks that guide the encoder. Temporal stability is achieved through TTL-based ROI persistence with IoU-based merge smoothing [15].

3.2 Risk Score Computation

The risk score R integrates multiple signals to produce a comprehensive threat assessment [16]. Each signal component is normalized to contribute to the $[0.0, 1.0]$ range:

$$R = \min(1.0, 0.30N_p + 0.10N_v + 0.25M + 0.20T + 0.15S) \quad (2)$$

The scene change magnitude S is computed using histogram comparison between consecutive frames:

$$S = \frac{1}{256} \sum_{i=0}^{255} |H_t(i) - H_{t-1}(i)| \quad (3)$$

where H_t is the normalized histogram of frame at time t .

3.3 Adaptive Encoding Strategy

Background frames and ROI crops are encoded separately as Base64 JPEG images. The data packet structure includes `frame_id`, `bg_data`, `rois` list with bounding boxes, `orig_w`, `orig_h`, `timestamp`, `risk_score`, and `surveillance_state`. Transmission uses Socket.IO for real-time streaming to the central server [20, 4].

3.4 Frame Storage and Reconstruction

Frames are stored in a structured directory format:

```
recordings/[date]/bg/      (background images)
recordings/[date]/roi/     (ROI crops)
recordings/[date]/meta.json (metadata)
events/                   (pre/post-event archives)
```

Stored frames can be reconstructed by loading the background frame and overlaying ROI images at their corresponding bounding box coordinates [21].

3.5 Privacy Operation Modes

Three privacy tiers address ethical surveillance concerns [22]:

- **Privacy Blur:** Heavy Gaussian background blur
- **Ethical Mode:** Black frame when no ROI detected (zero passive surveillance)
- **Face Masking:** Anonymization of detected persons while preserving vehicle visibility

4 Evaluation

4.1 Evaluation Pipeline

The automated evaluation pipeline iterates over configurable `{bg_quality, roi_quality}` presets, sends each as a live control command, and captures original/reconstructed frame pairs [23, 24, 25]. For each experiment, the following metrics are computed:

- Global PSNR and SSIM [23]
- Per-ROI PSNR and SSIM
- Background-only PSNR and SSIM
- VMAF perceptual quality scores
- Detection accuracy retention (YOLOv8 re-detection on compressed frames)

4.2 Bandwidth Reduction

The system achieves significant bandwidth savings through selective background compression [1, 15]:

Table 2: Bandwidth savings at different background scale factors

BG Scale	Avg. Bitrate	Savings	ROI PSNR (dB)
1× (full)	2000 kbps	—	38.2
2×	800 kbps	60%	36.8
4×	400 kbps	80%	35.1
8×	200 kbps	90%	32.4

4.3 Quality Metrics

Experimental results demonstrate stable quality metrics across compression levels:

Table 3: Quality metrics across compression presets

Preset	PSNR (dB)	SSIM	VMAF	Det. Accuracy
High Quality	36.2	0.95	92.4	98%
Balanced	33.8	0.91	87.1	95%
Ultra Low BW	30.1	0.85	78.5	88%
Privacy Mode	31.5	0.88	82.3	91%

4.4 Detection Performance

YOLOv8 nano achieves real-time inference on modest hardware [14]:

Table 4: YOLOv8 nano inference performance

Hardware	Inference (ms)	Max FPS
Raspberry Pi 4	120–150	6–8
Jetson Nano	30–50	20–30
x86 Laptop (CPU)	20–40	25–50
x86 Laptop (CUDA)	5–10	100+

5 Related Work

Recent advances in ROI-based video compression have demonstrated significant potential for bandwidth efficiency in surveillance systems [3]. Wang and Yang proposed a compression metadata-assisted ROI extraction framework for H.265/HEVC coding focused on edge-based video analytics, achieving latency reduction for detection tasks [3]. Dou et al. presented an R-lambda optimization based on CNN with ROI-based coding in the HEVC framework, achieving better perceptual quality in ROI regions with low bitrate overhead [4].

Labiod et al. proposed GMM-based ROI detection with cross-layer optimization for VANETs [6]. Guo et al. developed DeepStream, a bandwidth-efficient streaming architecture using YOLOv5-Lite with approximately 54% bandwidth reduction [5]. Gao et al. proposed a robust mixed-rate ROI-aware compressive sensing framework achieving PSNR of 34.71 dB [21].

Agrawal et al. proposed FRVC using YOLOv9 for frame relevance detection, achieving up to 85% storage reduction [15]. Gadot et al. proposed reinforcement learning-based rate control achieving 25% BD-rate reduction through block-level optimization [16]. Transformer-based approaches offer state-of-the-art compression but remain too computationally expensive for real-time edge deployment [26].

SDN-based adaptive streaming approaches provide dynamic bandwidth allocation across multiple cameras but lack content-aware ROI encoding [1]. Privacy-preserving techniques and federated learning approaches address important adjacent concerns but do not directly target compression efficiency [22, 27].

6 Conclusion

This paper presented an adaptive CCTV compression system that integrates YOLOv8-based dynamic ROI control with a risk-aware event engine to achieve intelligent bandwidth reduction. The three-tier architecture spanning edge node, central server, and interactive dashboard provides a complete, production-ready surveillance platform. Experimental results demonstrate 45–65% bandwidth reduction while maintaining stable perceptual quality metrics. The risk-aware event engine with pre/post-event frame buffering ensures forensic evidence preservation during critical incidents.

Future work includes deployment on dedicated edge AI hardware for improved inference performance, integration with cloud storage tiers, support for YOLOv8 segmentation for finer ROI boundaries, and multi-camera scalability validation.

Author Contributions

Conceptualization, H. Abdullah; methodology, S. I. Tamim, J. T. Nahar, M. M. Uddin; software, S. I. Tamim, J. T. Nahar, M. M. Uddin; validation, H. Abdullah; formal analysis, S. I. Tamim; investigation, J. T. Nahar; resources, M. M. Uddin; data curation, S. I. Tamim; writing—original draft preparation, S. I. Tamim, J. T. Nahar, M. M. Uddin; writing—review and editing, H. Abdullah; visualization, J. T. Nahar; supervision, H. Abdullah.

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